

Appendix

Rule 1 (tensor shapes should be the same)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}\} \models \text{ndim}(v_1) = \text{ndim}(v_2) \wedge \forall i \in [0, \text{ndim}(v_1) - 1] : \text{shape}(v_1, i) = \text{shape}(v_2, i)$$

Rule 2 (v_2 should be a valid dimension of input tensor)

$$\{v_1 : \text{tensor}, v_2 : \text{int}\} \models -1 \times \text{ndim}(v_1) \leq v_2 \wedge v_2 \leq \text{ndim}(v_1) - 1$$

Rule 3 (input tensors must have the same number of dimensions)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}\} \models \text{ndim}(v_1) = \text{ndim}(v_2)$$

Rule 4 (tensor data types should match)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}\} \models \text{dtype}(v_1) = \text{dtype}(v_2)$$

Rule 5 (index tensor (v_2) should be within the range of input tensor)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}\} \models \min(v_2) \geq 0 \wedge \forall i \in [0, \text{ndim}(v_1) - 1] : \max(v_2) \leq \text{shape}(v_1, i) - 1$$

Rule 6 (last dimension of a tensor matches first dimension of the other)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}\} \models \text{ndim}(v_1) > 0 \wedge \text{ndim}(v_2) > 0 \wedge \text{shape}(v_1, \text{ndim}(v_1) - 1) = \text{shape}(v_2, 0)$$

Rule 7 (first dimension of a tensor matches second dimension of the other)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}\} \models \text{ndim}(v_1) \geq 1 \wedge \text{ndim}(v_2) \geq 2 \wedge \text{shape}(v_1, 0) = \text{shape}(v_2, 1)$$

Rule 8 (tensor must have an integer dtype: 1-5)

$$\{v_1 : \text{tensor}\} \models 1 \leq \text{dtype}(v_1) \leq 5$$

Rule 9 (4D tensor with all shape dimensions positive)

$$\{v_1 : \text{tensor}\} \models \text{ndim}(v_1) = 4 \wedge \forall i \in [0, 3] : \text{shape}(v_1, i) > 0$$

Rule 10 (valid 2D transposed convolution)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}, v_3 : \text{int}, v_4 : \text{int}\} \models \text{ndim}(v_1) = 4 \wedge \text{ndim}(v_2) = 4 \wedge v_3 \geq 1 \wedge v_4 \geq 0 \wedge \\ v_3 \times (\text{shape}(v_1, 2) - 1) + \text{shape}(v_2, 2) - 2 \times v_4 > 0 \wedge v_3 \times (\text{shape}(v_1, 3) - 1) + \text{shape}(v_2, 3) - 2 \times v_4 > 0$$

Rule 11 (index tensor (v_3) is within the input tensor along the specified dimension)

$$\{v_1 : \text{tensor}, v_2 : \text{int}, v_3 : \text{tensor}\} \models \min(v_3) \geq 0 \wedge \max(v_3) \leq \text{shape}(v_1, v_2) - 1$$

Rule 12 (a primitive type variable is smaller than the other)

$$\{v_1 : \text{int} \sqcup \text{float}, v_2 : \text{int} \sqcup \text{float}\} \models v_1 \leq v_2$$

Rule 13 (tensor must have a floating-point dtype: 6-8)

$$\{v_1 : \text{tensor}\} \models 6 \leq \text{dtype}(v_1) \leq 8$$

Rule 14 (tensor should not be empty)

$$\{v_1 : \text{tensor}\} \models \forall i \in [0, \text{ndim}(v_1) - 1] : \text{shape}(v_1, i) > 0$$

Rule 15 (broadcasting compatibility for tensor shapes)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}\} \models \text{ndim}(v_1) > 0 \wedge \text{ndim}(v_2) > 0 \wedge \\ \forall i \in [0, \text{if } \text{ndim}(v_1) \geq \text{ndim}(v_2) \text{ then } \text{ndim}(v_1) - 1 \text{ else } \text{ndim}(v_2) - 1] : \text{ndim}(v_1) - i - 1 < 0 \vee \text{ndim}(v_2) - i - 1 < 0 \vee \\ \text{shape}(v_1, \text{ndim}(v_1) - i - 1) = 1 \vee \text{shape}(v_2, \text{ndim}(v_2) - i - 1) = 1 \vee \text{shape}(v_1, \text{ndim}(v_1) - i - 1) = \text{shape}(v_2, \text{ndim}(v_2) - i - 1)$$

Rule 16 (shape alignment for matrix multiplication)

$$\{v_1 : \text{tensor}, v_2 : \text{tensor}\} \models (\text{ndim}(v_1) \geq 2 \wedge \text{ndim}(v_2) \geq 2 \wedge \text{shape}(v_1, \text{ndim}(v_1) - 1) = \text{shape}(v_2, \text{ndim}(v_2) - 2)) \vee \\ (\text{ndim}(v_1) \geq 2 \wedge \text{ndim}(v_2) = 1 \wedge \text{shape}(v_1, \text{ndim}(v_1) - 1) = \text{shape}(v_2, 0))$$

Rule 17 (primitive type variable must be non-negative)

$$\{v_1 : \text{int} \sqcup \text{float}\} \models v_1 \geq 0$$

Rule 18 (input tensor is empty or only contains non-negative values)

$$\{v_1 : \text{tensor}\} \models (\exists i \in [0, \text{ndim}(v_1) - 1] : \text{shape}(v_1, i) = 0) \vee \min(v_1) \geq 0$$

Rule 19 (compatibility of integer or float (data) type tensor and variable)

$$\{v_1 : \text{tensor}, v_2 : \text{int} \sqcup \text{float}\} \models \text{if } 1 \leq \text{dtype}(v_1) \leq 5 \text{ then } 1 \leq v_2 \leq 5 \text{ else if } 6 \leq \text{dtype}(v_1) \leq 8 \text{ then } 1 \leq v_2 \leq 8 \text{ else false}$$

Rule 20 (input tensor must be 1-dimensional)

$$\{v_1 : \text{tensor}\} \models \text{ndim}(v_1) = 1$$

Rule 21 (primitive type variable should not be zero)

$$\{v_1 : \text{int} \sqcup \text{float}\} \models v_1 \neq 0$$

Rule 22 (valid step size for a range)

$$\{v_1 : \text{int} \sqcup \text{float}, v_2 : \text{int} \sqcup \text{float}, v_3 : \text{int} \sqcup \text{float}\} \models \text{if } v_1 > 0 \text{ then } v_2 \leq v_3 \text{ else if } v_1 < 0 \text{ then } v_2 \geq v_3 \text{ else true}$$

Rule 23 (3-dimensional tensor with all positive shape dimensions)

$$\{v_1 : \text{tensor}\} \models \text{ndim}(v_1) = 3 \wedge \forall i \in [0, \text{ndim}(v_1) - 1] : \text{shape}(v_1, i) > 0$$

Figure 1: DL Library API Constraints in Formal Grammar